

HAMMADUR RAHMAN

AI Engineer — GenAI Agents · RAG · Python · FastAPI · Docker

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New Delhi, India | Open to relocation (Bangalore / Chennai) | Notice period: 15 days

SUMMARY

Applied AI Engineer with 1+ year building and shipping **production LLM systems** — agents, RAG pipelines, multi-stage pipeline architectures, and evaluation harnesses — into live business workflows. Architected a multi-stage LLM pipeline for an enterprise workforce-planning platform, combining structured action-sequence modeling, financial-reporting automation, and a **reinforcement-learning** training environment; separately delivered a document-QA platform running at **90%+ answer accuracy** in daily production use. Python, LangChain / LangGraph, FastAPI, Docker, GCP.

PROFESSIONAL EXPERIENCE

AI Engineer / Data Scientist

June 2025 – Present

Blackcoffer

Remote / New Delhi

- Architected the platform's **OKR/KPI cascade** (company objectives, KPI-to-department routing, department-level OKR generation — live-tested end-to-end) and a **deterministic cost/duration reconciliation engine** (task-action sequences to auditable cost model, $\pm 15\%$ tolerance), both backed by **deterministic Python validators** layered beneath LLM output to catch self-audit failures the model missed on its own.
- Built a **financial-reporting automation API** mapping operational activity data to **US-GAAP-aligned** statement line items (balance sheet / income statement, cited accounting references) and a **deterministic Unity ML-Agents compiler** (PPO, ONNX) — a dual-output redesign cutting payload 60%+, verified via a 32-check harness against a partner team's integration spec.
- Built and shipped **LLM agents and RAG pipelines** (LangGraph, LangChain, OpenAI GPT-4, ChromaDB) into live client knowledge workflows, reaching **90%+ answer accuracy** through retrieval tuning, prompt engineering, **guardrails**, and systematic **response evaluation**; standardised agent behaviour with reusable system rules reused across internal and client-facing systems.
- Exposed these systems as **FastAPI REST services** (async, middleware, JSON), containerised with **Docker** and deployed via **GitHub Actions CI/CD** — sustaining **99%+ uptime across 1,000+ daily requests** with structured logging, output-quality monitoring, and threshold-based alerting.
- Engineered a **multi-source ingestion layer** integrating the Xandr, Google AdSense, and AdPushup **APIs** into a **BigQuery** warehouse (GCP), processing **1.79M+ records** and surfacing live MIS dashboards in Looker Studio adopted by stakeholders for weekly reporting; **led a sub-team of two** and drove tool adoption through documentation and hands-on stakeholder enablement.

PROJECTS (GitHub: VnMxMadMax)

Sentinel AI — Multi-Agent Monitoring & Automated Incident Response

Python, LangGraph, FastAPI, WebSockets, OpenAI API, ChromaDB, LangSmith, Docker

- Multi-agent system that detects cascading failure signals in a monitored service and drafts remediation, gated by an **automated evaluation harness** that scores agent output before it is acted on; includes **cost guardrails** on LLM calls and **LangSmith tracing** for step-level debugging. Streams real-time updates over **WebSockets**, served via FastAPI and containerised with Docker.

Production RAG System — Document Q&A over Multi-Store Retrieval

Python, LangChain, OpenAI API, Pinecone, FAISS, ChromaDB, Neo4j (GraphRAG), FastAPI

- Retrieval-augmented Q&A pipeline indexing **10,000+ document chunks**, **benchmarked across four retrieval backends** (Pinecone, FAISS, ChromaDB, and Neo4j GraphRAG) to compare retrieval quality on the same corpus; built a **retrieval evaluation set** and **hallucination-aware reliability testing** to catch ungrounded answers, served as a FastAPI REST API.

Fraud Detection Service — Imbalanced-Data Classification

Python, XGBoost, Scikit-learn, Pandas, NumPy, SHAP, Flask, Docker

- **End-to-end ML pipeline** — data preparation, feature engineering, training, evaluation, deployment — for **XGBoost fraud classification** on highly imbalanced data, achieving **94% precision**; served over a Flask REST endpoint with **SHAP explainability**, model versioning, and Docker deployment.

TECHNICAL SKILLS

- **Languages:** Python (advanced) · SQL (PostgreSQL, BigQuery, MySQL) · Java (fundamentals) · Git / GitHub
- **GenAI & Agents:** LangChain · LangGraph · OpenAI GPT-4 · Anthropic Claude · Google Gemini · Hugging Face Transformers · CrewAI · AutoGen · MCP (Model Context Protocol) · tool-calling / function-calling · multi-agent workflows · prompt engineering · guardrails · LLM evaluation & hallucination mitigation
- **RAG & Vector Stores:** RAG architectures · GraphRAG · embeddings, tokenization & chunking · Pinecone · FAISS · ChromaDB · Neo4j · retrieval tuning & evaluation
- **ML & Data:** Scikit-learn · XGBoost · Pandas · NumPy · classification & predictive modeling · imbalanced-data methods · SHAP explainability · NLP · feature engineering
- **Engineering & Ops:** FastAPI · Flask · REST API design · microservices · async / await · WebSockets · webhooks · Docker · GitHub Actions CI/CD · GCP (BigQuery, Looker Studio) · AWS · Pytest · structured logging & monitoring

EDUCATION

B.Tech, Computer Science — CGPA 7.7 / 10

Delhi Technical Campus (GGSIPU)

Sept 2021 – July 2025

New Delhi, India

CERTIFICATIONS & ACHIEVEMENTS

- **IBM (Coursera):** RAG and Agentic AI — Professional Certificate ([link](#)) · Agentic AI with LangGraph, CrewAI, AutoGen and BeeAI ([link](#)) · Build Multimodal Generative AI Applications ([link](#))
- **200+ DSA problems solved** on [GeeksforGeeks](#)